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The canary in the coal decline: Appalachian household finance and the transition from fossil fuels

Contents

1 Big picture

- **Question.** How did the recent decline in Appalachian coal mining affect household finances, and how quickly did the costs appear?
- **Setting.** The paper studies Appalachian coal counties from 2011 to 2018, a period when Appalachian coal production fell by about 40 percent and coal employment fell by about 49 percent.
- **Main data.** The core household finance data are the New York Fed/Equifax Consumer Credit Panel, a 5 percent random sample of US individuals with credit reports, linked to county-level coal demand and production measures.
- **Treatment.** The main treatment is annual coal consumed at power plants within 200 miles of a county by rail. It captures local electricity-sector demand for Appalachian coal.
- **Identification idea.** The decline in coal demand was driven by low natural gas prices and preexisting natural gas power capacity, not by household financial conditions in Appalachian coal counties.
- **Main result.** The average annual coal demand decline lowered credit scores, raised subprime status, increased credit utilization, increased delinquent accounts, raised third-party collections, and raised bankruptcy transitions.
- **Magnitude.** The baseline credit-score coefficient is about -0.386 points per year. Multiplied by the seven-year sample window, the implied net credit-score effect is about -2.7 points; around the subprime threshold, quantile effects are more than twice as large.
- **Timing.** Distributed lags and local projections show that credit outcomes deteriorate quickly, generally within two years of coal demand shocks.
- **Distribution.** Effects are strongest for lower-middle credit-score households. The 40th percentile is especially affected, near the subprime cutoff where small score movements can change access to credit.
- **Age heterogeneity.** Baby Boomers and Generation X households are more affected than Millennials, consistent with longer local attachment and more labor-market exposure to coal communities.
- **Mechanism.** The effects are not explained solely by job losses in coal-worker households. Spillovers to the broader local economy appear important.
- **Migration.** The coal decline does not generate significant net migration, but it reduces churn and is associated with worse credit-score recovery for people who leave Appalachia during the decline period.
- **Takeaway.** Energy transitions can impose fast, financially meaningful, and uneven costs on fossil-fuel extraction communities, even when direct fossil-fuel employment is a small share of total employment.

2 Data and measurement

2.1 Coal mining counties

- The paper studies counties in the EIA-designated Appalachian coal mining basin.
- A county is an active coal-mining county if it has at least one year with nonzero coal production and at least one year with at least 10,000 annual employee hours in mining during 2011–2018.
- The sample drops Allegheny County, Pennsylvania, because Pittsburgh makes it unrepresentative of the Appalachian coal region.
- Average active coal-mining county production is about 2.4 million short tons per year, with roughly one million employee hours and about 450 coal employees.

Interpretation: Coal was important to local economies, but direct coal employment was small relative to total county employment. This makes the household finance effects especially informative about spillovers beyond coal workers.

2.2 Household finance outcomes

- The household finance data come from the NY Fed/Equifax Consumer Credit Panel.
- The sample is an individual-level annual panel from 2011 to 2018, using fourth-quarter outcomes.
- Individuals must be 25–64 years old when entering the sample.
- Main outcomes are credit score, subprime status, credit utilization, delinquent accounts, third-party collections, and bankruptcy entry.
- The credit-score variable is the Equifax Risk Score, which is similar to other credit scores used in consumer lending.
- Subprime equals one when the credit score is below 660.
- Delinquent accounts count accounts that are 60, 90, 120, or more days past due.
- Third-party collections capture debt owed to collection agencies, including debts that may not appear in standard delinquency measures.
- Bankruptcy equals one if the individual enters Chapter 7 or Chapter 13 bankruptcy in a year.

Interpretation: These outcomes measure both current financial distress and future credit access. A small credit-score decline can be economically meaningful when it moves a borrower close to lending thresholds.

2.3 Coal demand measure

- The paper constructs a local coal-demand variable from coal consumed at power plants within 200 miles of a county centroid by rail.
- Coal transportation costs make geographic proximity important: power plants near Appalachian mines are the relevant demand shifters.
- The decline in coal consumption is tied to low natural gas prices and the ability of utilities to switch generation from coal to gas.
- Cross-sectional variation comes partly from pre-2009 natural gas generation capacity, which was predetermined relative to the 2011–2018 coal decline.
- The treatment is rescaled by the average annual decline in coal demand over 2011–2018, about 4.1 million tons per year.

Interpretation: The empirical effect is interpreted as the average one-year change in a household finance outcome caused by the average annual decline in local coal demand during the sample period.

2.4 Summary statistics

Read:

- The national sample has an average credit score around 686, while active coal-mining counties average around 678.
- Active coal-mining counties have higher subprime shares than the national sample.
- Credit utilization, delinquent accounts, third-party collections, bankruptcy, and age are similar in broad levels, but coal counties have weaker credit-score distributions.

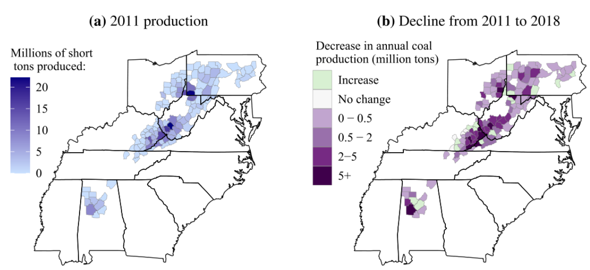
Economic message: The coal counties start from a more fragile financial position, so a coal-demand shock can have larger consequences for borrowing access.

Table 1
Summary statistics.

	Active Coal Mining Counties	National Sample
Coal demand and production		
Coal consumed w/in 200 mi by rail (mil short tons)	31.47	.
	(20.44)	
Total annual coal production (mil. tons)	2.43	.
	(3.82)	
Total annual employee hours worked (Thous.)	1,017.86	.
	(1,404.89)	
Number of employees	450.56	.
	(606.50)	
FRB New York / Equifax CCP data		
Credit score	678.46	686.13
	(109.06)	(106.88)
Percent subprime	41.41	38.50
	(49.26)	(48.66)
Credit utilization (percent)	39.79	38.93
	(39.05)	(38.64)
Delinquent accounts	0.12	0.12
	(0.71)	(0.66)
Amt 3rd party (2012 $)	251.59	233.10
	(1,716.07)	(2,006.37)
Transition to bankruptcy (percent)	0.39	0.37
	(6.24)	(6.05)
Age	44.58	44.68
	(11.48)	(11.44)

Source: EIA; Census; NY Fed / Equifax CCP
Note: Standard deviation in parentheses. We define "active coal-mining counties" during 2011-2018 as those with (i) at least one year with non-zero coal production and (ii) at least one year with 10,000 hours or more of total annual employee hours in mining (roughly equivalent to five full-time workers). Credit score is the fourth quarter value. Subprime is defined as having a credit score below 660. Credit utilization percent is based on bankcard and retail trade balances and credit limits. Delinquent accounts is the total number of accounts that are 60, 90, 120 or more days past due in the fourth quarter. Amount in 3rd party collections reflects the fourth quarter amount. Bankruptcy equals 1 if in-

(a) Table 1: summary statistics



l production in Appalachia
own are active Appalachian coal mining counties between 2011 and 2018, our sample period. Defined as: (i) at least one year with non-zero and at least one year with 10,000 h or more of total annual employee hours in mining (roughly equivalent to five full-time workers) according to rmation Admin- istration (EIA) and (ii) U.S. Mine Safety and Health Administration (MSHA). Counties are considered in Appalachia if they are within ted Appalachian mining basins.

(b) Figure 1: Appalachian coal production and decline

3 Empirical design and specifications

3.1 Baseline individual panel regression

The main specification is

$$Y_{ict} = \beta_1 + \beta_2 Coal200_{ct} + \delta_i t + \alpha_i + \eta_c + \mu_t + \varepsilon_{ict}. \tag{1}$$

- Y_{ict} is the outcome for individual i in county c and year t.
- $Coal200_{ct}$ is coal burned at power plants within 200 miles of county c by rail.
- α_i are individual fixed effects.
- η_c are county fixed effects.
- μ_t are year fixed effects.
- $\delta_i t$ is an individual-specific linear time trend.
- Standard errors are clustered by county.

Interpretation: The coefficient β_2 compares changes in credit outcomes of people in counties experiencing larger coal-demand declines to people in counties experiencing smaller declines, after controlling for individual, county, year, and individual-trend differences.

3.2 Validation regression for coal production

The paper validates the demand measure with a county-level regression:

$$Y_{ct} = \gamma_1 + \gamma_2 Coal200_{ct} + \eta_c t + \nu_c + \mu_t + \varepsilon_{ct}, \tag{2}$$

where Y_{ct} is coal production, mining hours, or coal employment.

Interpretation: If coal consumed at nearby power plants is a relevant local demand shock, it should predict coal production and mining employment in nearby coal counties. Table 2 shows that it does.

3.3 Distributed lag specification

To study timing, the paper adds lags of coal demand:

$$Y_{ict} = \beta_0 Coal200_{ct} + \beta_1 Coal200_{c,t-1} + \beta_2 Coal200_{c,t-2} + \text{fixed effects and trends} + u_{ict}. \tag{3}$$

The paper also reports cumulative effects across the contemporaneous, one-year-lag, and two-year-lag coefficients.

Interpretation: The distributed lag model asks whether household finance deteriorates immediately or only after coal demand shocks work through production, employment, and local labor markets.

3.4 Local projections

The local projection specification is

$$Y_{i,c,t+h} - Y_{i,c,t-1} = \beta^h \Delta coal200_{c,t} + \sum_{\tau=-3,\tau\neq 0}^{h} \delta_\tau^h \Delta coal200_{c,t+\tau} + \sum_{\tau=-3}^{-1} \rho_\tau^h \Delta y_{i,t-\tau} + \alpha_i^h + \alpha_c^h + \alpha_t^h + \varepsilon_{c,t+h}. \tag{4}$$

Interpretation: The local projection coefficients trace impulse responses of credit outcomes over horizons after a coal-demand shock. This directly shows whether the effect is temporary, delayed, or persistent.

3.5 Heterogeneity specifications

The paper modifies the baseline regression by interacting coal demand with:

- pre-period credit-score quartiles;
- credit-score quantiles using unconditional quantile regressions;
- generation indicators and credit-score bins;
- Census block group poverty shares;
- income and employment controls;
- migration status and 2010 county of residence.

Interpretation: These specifications ask who bears the costs and whether effects arise from direct coal jobs, broader local spillovers, household fragility, or migration-driven sample composition.

4 Identification and empirical logic

4.1 Why the coal-demand shock is useful

- Low natural gas prices after the fracking boom led utilities to substitute away from coal.
- Counties near power plants with more ability to switch to natural gas experienced larger declines in demand for nearby Appalachian coal.
- Preexisting gas capacity was largely built before the 2011–2018 coal decline and is therefore plausibly unrelated to contemporaneous household credit shocks.
- The use of rail distance ties demand shocks to economically relevant coal transportation markets.

Interpretation: The treatment is not county-level coal production itself. It is electricity-sector demand for nearby coal, which is less likely to be caused by local household financial conditions.

4.2 Reduced form rather than IV

- The coal-demand variable predicts coal production, hours, and employment, but it is not strong enough to serve as a conventional instrument for county-level coal production in the main individual-level regression.
- The paper therefore estimates reduced-form effects of demand for nearby coal on household finance.

Interpretation: The advantage is avoiding weak-IV bias. The tradeoff is that the estimated effect is interpreted per change in coal demand, not per change in actual local coal production.

4.3 Main identifying assumption

- Conditional on fixed effects and individual time trends, deviations in coal demand are exogenous and unanticipated with respect to household credit outcomes.
- Individuals in counties with small coal-demand changes provide the counterfactual for individuals in counties with large coal-demand declines.
- The causal response to a dose of coal-demand decline is assumed comparable across counties in the same time period.

Interpretation: This is a continuous-treatment fixed-effects design, not a sharp event-study design with treated and untreated cohorts.

4.4 Threats and responses

- **Local confounds.** Alternative radii, excluding nearby coal consumption, and controlling for income/employment reduce concern that results are driven by unrelated local trends.
- **Migration.** Results survive assigning treatment based on 2010 county and restricting to non-movers.
- **Timing.** Effects emerge within two years, consistent with the timing of coal-demand shocks.
- **Direct job losses.** Back-of-the-envelope calculations show direct coal-worker job losses are too small to explain the full sample-wide credit-score decline.

Interpretation: The paper's strongest evidence is not any single regression. It is the alignment between the demand shock, coal-production validation, baseline credit outcomes, dynamics, heterogeneity, and migration robustness.

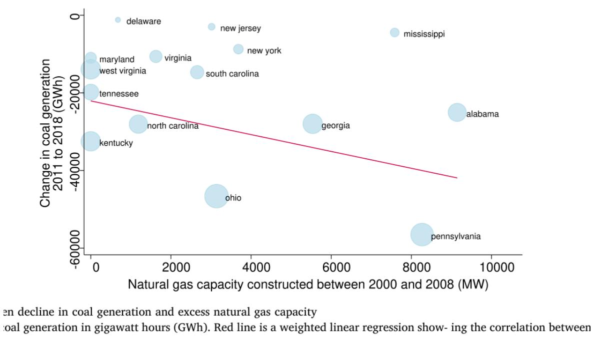
5 Tables and figures

5.1 Figures 2–3: why nearby coal demand changes

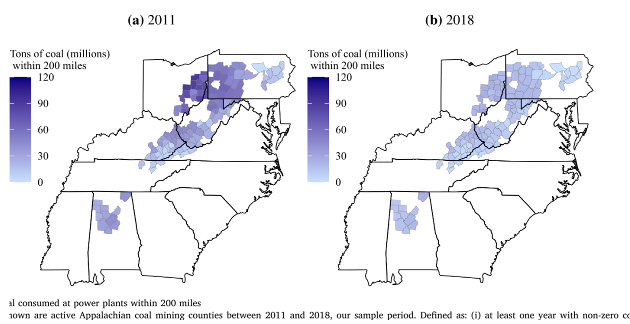
Read:

- Figure 2 shows that states with more natural gas capacity built before 2009 experienced larger coal generation declines from 2011 to 2018.
- The point sizes represent 2007 coal generation, so the relation is economically relevant for coal-heavy states.
- Figure 3 maps coal consumed at power plants within 200 miles of active coal-mining counties in 2011 and 2018.
- Coal demand falls substantially across Appalachian coal counties over the sample period.

Economic message: The figures motivate the shock: utilities could switch from coal to gas when gas became cheap, and this generated spatially uneven demand declines for Appalachian coal.



(a) Figure 2: gas capacity and coal generation decline



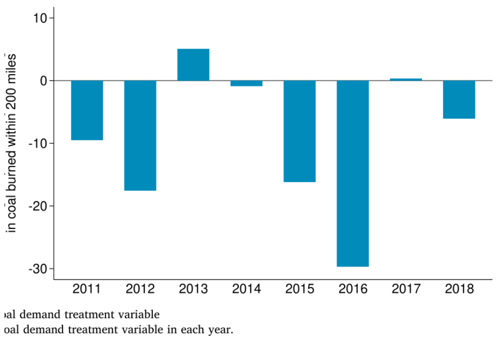
(b) Figure 3: coal consumed within 200 miles

5.2 Figure 4 and Table 2: treatment timing and validation

Read:

- Figure 4 shows negative coal-demand changes in most years, with especially large declines in 2012, 2015, 2016, and 2018.
- Table 2 shows that a decline in coal consumed within 200 miles predicts lower coal production, mining hours, and coal employment.
- The coal production coefficient is -0.116 million tons, the hours coefficient is -60.80 thousand, and the employment coefficient is -30.46 employees.

Economic message: The treatment variable is not just an abstract power-sector measure. It is statistically linked to actual coal mining activity in Appalachian counties.



(a) Figure 4: year-to-year coal demand variation

Relationship between coal tons demanded and coal production measures.

	(1) Coal Production (millions of tons)	(2) Hours (thousands)	(3) Employees (count)
Coal tons 200m	−0.116***	−60.80***	−30.46***
	(−2.72)	(−3.37)	(−3.02)
Mean of Dep. Var.	2.02	773.11	412.88
Observations	976	976	822
Number of Counties	122	122	119

Source: MSHA; EIA; NY Fed / Equifax CCP
Notes: Observations are at county-year level. Coal tons 200m captures millions of coal tons burned within 200 miles of a county centroid by rail. To help with interpretation, coal tons 200m is rescaled by the aver- age annual change between 2011 and 2018. Treatment coefficients are the average one-year change in the outcome variable caused by the coal decline between 2011 and 2018. The sample spans 2011-2018. Regressions are weighted by each county's 2007-2010 coal production, and include county and year fixed effects as well as county time trends. Standard errors are clustered by county. *, **, and *** represent significance at the 10, 5, and 1 percent levels, respectively.

(b) Table 2: coal demand and production measures

5.3 Table 3 and Figure 5: baseline household finance effects

Read:

- Table 3 shows that the average annual coal demand decline lowers credit scores by 0.386 points.
- The same shock raises subprime status by 0.233 percentage points, credit utilization by 0.187 percentage points, delinquent accounts by 0.00307, third-party collections by about 9 dollars, and bankruptcy by 0.0391 percentage points.
- Multiplying by seven years implies an average net credit-score effect of roughly 2.7 points.
- Figure 5 compares observed outcome paths to 2018 counterfactuals if coal demand had stayed at 2011 levels.

Economic message: The coal decline worsens household balance sheets along multiple dimensions. The effects are modest on average but meaningful because credit outcomes are threshold-sensitive.

Table 3
Credit outcomes.

	(1) Credit Score	(2) Percent Subprime	(3) Credit Utilization Percent	(4) Delinquent Accounts	(5) Amt 3rd Party Collections ($ 2012)	(6) Percent Bankruptcy
Coal tons 200m	−0.386***	0.233***	0.187***	0.00307**	9.034***	0.0391***
	(−2.79)	(3.98)	(4.39)	(2.40)	(2.83)	(3.40)
Mean of Dep. Var.	680.54	41.02	38.70	0.11	206.63	0.47
Observations	1,498,042	1,498,042	1,078,856	1,518,432	1,545,450	1,597,776
Individuals	228,128	228,128	179,861	221,266	232,841	234,259
Individ w/ Outcome	228,128	124,905	174,808	51,288	101,029	7,523

Source: EIA; NY Fed / Equifax CCP
Note: Coal tons 200m captures coal (millions of tons) burned within 200 miles of a county centroid by rail. To help with interpretation, coal tons 200m is rescaled by the average annual change between 2011 and 2018. Treatment coefficients are the average one-year change in the outcome variable caused by the coal decline between 2011 and 2018. Observations are at individual-year level and span 2011-2018. Subprime equals 1 when credit score is below 660. Credit utilization percent is based on bankcard and retail trades. Delinquent accounts is the total number of accounts that are 60, 90, 120 or more days past due. Bankruptcy equals 1 when

Figure 4: Table 3: baseline effects on credit outcomes

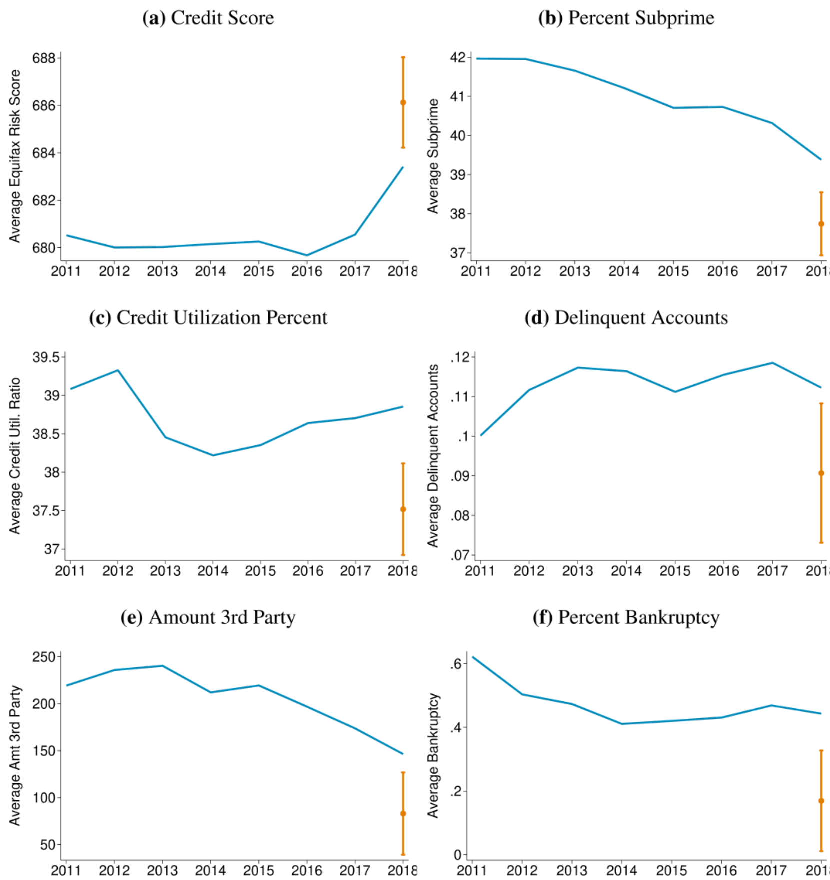


Figure 5: Figure 5: counterfactual 2018 outcomes absent the coal demand decline

5.4 Tables 4–5: robustness to coal-demand radius and sample period

Read:

- Table 4 excludes coal burned within 50 miles and uses coal burned between 50 and 200 miles; the baseline pattern remains.
- Table 5 extends the sample and also creates a gap between 2002–2007 and 2011–2018.
- The full-period and gapped samples produce credit-score declines similar to or larger than the baseline.
- In 2002–2007 alone, when coal demand was not declining significantly, the credit-score coefficient is not statistically significant.

Economic message: The main findings are not driven by a mechanical relationship with very nearby power plants or by generic Appalachian credit trends in earlier periods.

gesting that a larger concentration of coal mine workers results in larger levels of natural gas production show higher costs of the coal decline.

Table 4
Coal demand measured as coal burned within 50-200 miles.

	(1) Credit Score	(2) Percent Subprime	(3) Credit Utilization Percent	(4) Delinquent Accounts	(5) Amt 3rd Party Collections ($ 2012)	(6) Percent Bankruptcy
Coal tons 50m-200m	−0.352**	0.208***	0.161***	0.00290**	8.006***	0.0314***
	(−2.59)	(3.62)	(3.63)	(2.26)	(2.69)	(2.78)
Mean of Dep. Var.	680.54	41.02	38.70	0.11	206.63	0.47
Observations	1,498,042	1,498,042	1,078,856	1,518,432	1,545,450	1,597,776
Individuals	228,128	228,128	179,861	221,266	232,841	234,259
Individ w/ Outcome	228,128	125,781	174,995	53,137	102,384	7,877

Source: EIA; NY Fed / Equifax CCP
Notes: Coal tons 50m-200m captures millions of coal tons burned between 50-200 miles of a county's centroid by rail. To help with interpretation, coal tons 200m is rescaled by the average annual change between 2011 and 2018. Treatment coefficients are the average one-year change in the outcome variable caused by the coal decline between 2011 and 2018. Subprime equals 1 when credit score is below 660. Credit utilization percent is based on bankcard and retail trades. Delinquent accounts is the total number of accounts that are 60, 90, 120 or more days past due. Bankruptcy equals 1 when individuals transition into chapter 7 or 13 bankruptcy. Years included are 2011-2018. All regressions include individual, county, and year FE as well as linear individual time trends. Standard errors clustered at the county level. *, **, and *** represent significance at the 10, 5, and 1 percent levels, respectively.

Figure 6: Table 4: coal demand measured as coal burned within 50–200 miles

Table 5
Alternative time periods.

(a) Full time period (2002-2018)

	(1) Credit Score	(2) Percent Subprime	(3) Credit Utilization Percent	(4) Delinquent Accounts	(5) Amt 3rd Party Collections ($ 2012)	(6) Percent Bankruptcy
Coal tons 200m	−0.444***	0.204***	0.102**	0.00161	6.086*	0.0293***
	(−3.46)	(4.00)	(2.43)	(1.41)	(1.97)	(2.99)
Mean of Dep. Var.	677.16	41.72	42.24	0.12	223.55	0.71
Observations	3,383,302	3,383,302	2,551,169	3,343,974	3,478,857	3,340,346
Individuals	311,073	311,073	264,542	293,237	313,592	306,467
Individ w/ Outcome	311,073	192,847	258,752	104,711	154,857	26,633

(b) Time period with gap (2002-2007, 2011-2018)

	(1) Credit Score	(2) Percent Subprime	(3) Credit Utilization Percent	(4) Delinquent Accounts	(5) Amt 3rd Party Collections ($ 2012)	(6) Percent Bankruptcy
Coal tons 200m	−0.461**	0.225***	0.0933*	0.00135	5.576	0.0209*
	(−2.60)	(3.31)	(1.76)	(1.08)	(1.54)	(1.97)
Mean of Dep. Var.	676.90	41.62	42.27	0.11	216.85	0.69
Observations	2,746,657	2,746,657	2,075,121	2,724,395	2,822,822	2,678,608
Individuals	303,505	303,505	256,211	286,909	306,387	300,016
Individ w/ Outcome	303,505	185,301	250,568	92,650	145,505	20,443

(c) 2002-2007

	(1) Credit Score	(2) Percent Subprime	(3) Credit Utilization Percent	(4) Delinquent Accounts	(5) Amt 3rd Party Collections ($ 2012)	(6) Percent Bankruptcy
Coal tons 200m	−0.300	−0.0438	−0.126	0.00497**	0.816	−0.00749
	(−1.40)	(−0.54)	(−1.33)	(2.50)	(0.09)	(−0.21)
Mean of Dep. Var.	674.01	41.72	45.78	0.12	180.26	1.04
Observations	1,162,537	1,162,537	927,039	1,124,837	1,188,219	994,488
Individuals	214,094	214,094	181,907	203,374	217,017	211,047
Individ w/ Outcome	214,094	114,152	176,586	45,877	78,276	10,451

Source: Census; EIA; NY Fed / Equifax CCP
Notes: Coal tons 200m captures millions of coal tons burned within 200 miles of a county centroid by rail. To help with interpretation, coal tons 200m is rescaled by the average annual change between 2011 and 2018. Treatment coefficients are the average one-year change in the outcome variable caused by the coal decline between

Figure 7: Table 5: alternative time periods

5.5 Table 6 and Figure 6: dynamics

Read:

- Table 6 decomposes the effect into contemporaneous, one-year lag, and two-year lag components.
- Credit scores fall contemporaneously and again after one year; the two-year lag is small.
- The cumulative credit-score effect is -0.455 points per annual shock; the cumulative subprime and credit-utilization effects are also statistically significant.
- Figure 6 local projections show that effects generally appear immediately or within two years and persist for several years, although longer-horizon confidence intervals widen.

Economic message: Household financial distress materializes quickly. This matters for policy because delayed assistance may miss the period when credit outcomes first deteriorate.

Table 6
Coal demand distributed lag model.

	(1) Credit Score	(2) Percent Subprime	(3) Credit Utilization Percent	(4) Delinquent Accounts	(5) Amt 3rd Party Collections ($ 2012)	(6) Percent Bankruptcy
Coal tons 200m	−0.277**	0.146***	0.181***	0.00381***	6.063**	0.0335**
	(−2.37)	(3.11)	(3.86)	(2.86)	(2.26)	(2.40)
Lag Coal tons 200m	−0.222*	0.134***	0.0139	−0.000426	2.011	0.000403
	(−1.97)	(2.62)	(0.32)	(−0.33)	(0.69)	(0.03)
2nd Lag Coal tons 200m	0.0441	0.0424	0.0465	−0.00115	2.860	0.0186
	(0.49)	(0.94)	(1.12)	(−1.03)	(1.18)	(1.43)
Cumulative effect	-0.455**	0.322***	0.242***	0.002	10.934***	0.053***
SE	-2.27	3.70	4.13	1.49	2.71	3.42
Mean of Dep. Var.	683.92	39.93	38.16	0.11	198.36	0.47
Observations	1,393,367	1,393,367	1,011,513	1,414,064	1,436,379	1,520,230
Individuals	217,911	217,911	171,521	212,520	222,858	228,849
Individ w/ Outcome	217,911	119,395	167,155	51,648	98,561	7,788

Source: Census; EIA; NY Fed / Equifax CCP
Notes: Lag of coal demand is a 1 year lag. 2nd lag of coal demand is a 2 year lag. The cumulative effect estimates the net effect from the contemporaneous, one-year lag, and two-year lag. Coal tons 200m captures coal (millions of tons) burned within 200 miles of a county centroid by rail. To help with interpretation, coal tons 200m is rescaled by the average annual change between 2011 and 2018. Treatment coefficients are the average one-year change in the outcome variable caused by the coal decline between 2011 and 2018. Observations are at individual-year level and span 2011-2018. Subprime equals 1 when credit score is below 660. Credit utilization percent is based on bankcard and retail trades. Delinquent accounts is the total number of accounts that are 60, 90, 120 or more days past due. Bankruptcy equals 1 when individuals transition into chapter 7 or 13 bankruptcy. All regressions include individual, county, and year fixed effects as well as linear individual time trends.

Figure 8: Table 6: coal demand distributed lag model

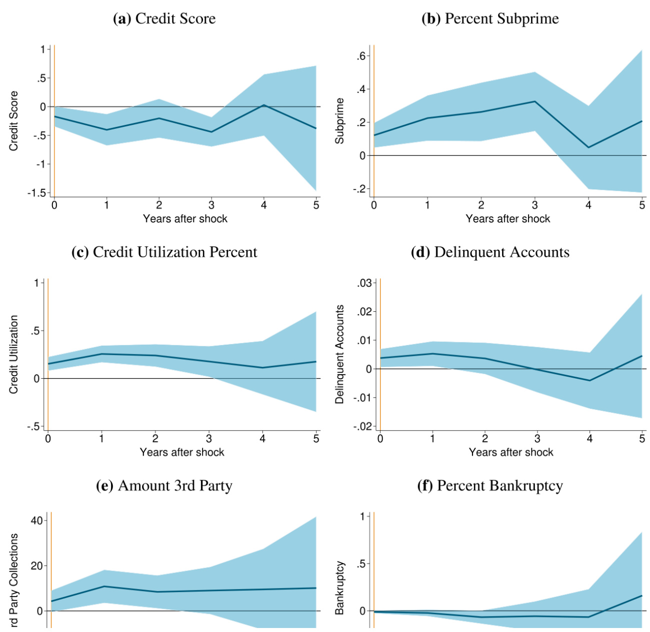


Figure 9: Figure 6: local projections

5.6 Table 7 and Figure 7: distributional effects by credit score

Read:

- Table 7 shows effects by pre-period credit-score quartile.
- Credit-score declines are statistically significant across quartiles, but financial distress outcomes are concentrated in the bottom half of the pre-period credit-score distribution.
- The second credit-score quartile sees the largest subprime increase and a large bankruptcy increase.
- Figure 7 shows quantile treatment effects: the largest credit-score decline appears near the 40th percentile, close to the subprime threshold.

Economic message: The most important financial costs fall near the point where a household can lose access to cheaper credit. Average effects understate the burden on households close to lending cutoffs.

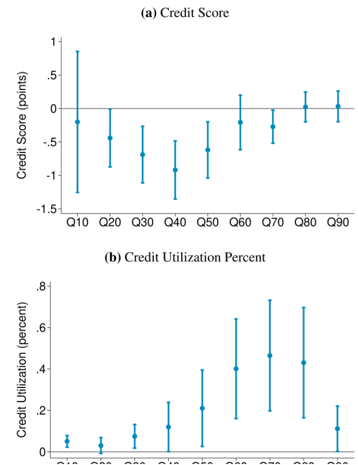
which translates to a net 7-point decline between 2011–2018. contrast, the top and very bottom ends of the distribution do not shi all. For credit utilization (panel (b)), the responses in the middle-up

Table 7
Coal demand interacted with baseline credit score quartile.

	(1) Credit Score	(2) Percent Subprime	(3) Credit Utilization Percent	(4) Delinquent Accounts	(5) Amt 3rd Party Collections ($ 2012)	(6) Percent Bankruptcy
Coal * bottom ERS quart	−0.517*	0.361***	0.220	0.00524	18.47**	0.0751***
	(−1.79)	(2.90)	(1.01)	(1.43)	(2.13)	(2.70)
Coal * second ERS quart	−0.598**	0.428***	0.305***	0.00287	5.514	0.0871***
	(−2.35)	(3.02)	(2.86)	(1.25)	(1.47)	(3.33)
Coal * third ERS quart	−0.510***	0.190*	0.208***	0.00292*	0.619	0.00869
	(−2.93)	(1.85)	(3.32)	(1.96)	(0.45)	(0.41)
Coal * top ERS quart	−0.268**	0.102***	0.140***	0.000594	1.026	0.0123
	(−2.23)	(3.09)	(2.75)	(1.19)	(1.43)	(1.47)
Mean of Dep. Var.	686.36	39.15	37.98	0.11	193.08	0.48
Observations	1,338,990	1,338,990	985,549	1,356,272	1,371,606	1,445,728
Individuals	194,635	194,635	158,070	189,390	197,733	202,190
Individ w/ Outcome	194,635	102,862	154,061	45,124	84,529	7,312

Source: EIA; NY Fed / Equifax CCP
Notes: Coal demand within 200 miles is interacted with credit score quartiles. Individuals are placed in quartiles based on their median credit score between 2007 and 2010. Coal tons 200m captures coal (millions of tons) burned within 200 miles of a county centroid by rail. To help with interpretation, coal tons 200m is rescaled by the average annual change between 2011 and 2018. Treatment coefficients are the average one-year change in the outcome variable caused by the coal decline between 2011 and 2018. Observations are at individual-year level and span 2011-2018. Subprime equals 1 when credit score is below 660. Credit utilization percent is

(a) Table 7: coal demand interacted with baseline credit-score quartiles



(b) Figure 7: unconditional quantile regressions

5.7 Table 8: age and credit-score heterogeneity

Read:

- Table 8 interacts coal demand with generation and baseline credit-score bins.
- Baby Boomers and Generation X households experience the largest credit-score declines.
- Millennials show weaker and generally less precise effects, consistent with less exposure to coal careers and younger credit histories.
- Bankruptcy responses are concentrated among low-credit-score Baby Boomers and low-credit-score Generation X households.

Economic message: The costs are not only a function of initial credit score. Age and career attachment to the local coal economy also matter.

Table 8
Coal demand interacted with generations and credit score bins.

	(1) Credit Score	(2) Percent Subprime	(3) Credit Utilization Percent	(4) Delinquent Accounts	(5) Amt 3rd Party Collections ($ 2012)	(6) Percent Bankruptcy
Baby Boomers						
Coal* Low credit score	−0.562*	0.437**	0.294**	0.00173	6.123	0.0914***
	(−1.73)	(2.60)	(2.11)	(0.71)	(0.93)	(2.92)
Coal* High credit score	−0.270**	0.148**	0.150**	0.000492	0.411	−0.00303
	(−2.07)	(2.30)	(2.54)	(0.54)	(0.42)	(−0.19)
Generation X						
Coal* Low credit score	−0.574*	0.446***	0.222	0.00707***	13.13**	0.0788**
	(−1.97)	(2.97)	(1.24)	(2.77)	(2.16)	(2.48)
Coal* High credit score	−0.576***	0.193*	0.220***	0.00301**	0.838	0.0365
	(−3.38)	(1.97)	(3.77)	(2.30)	(0.55)	(1.43)
Millennials						
Coal* Low credit score	−0.164	0.197	0.343	−0.000844	23.55**	0.0628
	(−0.36)	(0.89)	(1.21)	(−0.12)	(2.48)	(1.54)
Coal* High credit score	−0.387	0.208	0.129	0.00492	4.733	−0.0176
	(−1.33)	(1.29)	(0.90)	(1.49)	(1.40)	(−0.44)
Mean of Dep. Var.	686.36	39.15	37.98	0.11	193.08	0.48
Observations	1,338,990	1,338,990	985,549	1,356,272	1,371,606	1,445,728
Individuals	194,635	194,635	158,070	189,390	197,733	202,190
Individ w/ Outcome	194,635	102,862	154,061	45,124	84,529	7,312

Source: EIA; NY Fed / Equifax CCP
Notes: Coal demand within 200 miles is interacted with (1) generations and (2) credit score bins. To help with interpretation, coal tons 200m is rescaled by the average

Figure 11: Table 8: coal demand interacted with generation and credit-score bins

5.8 Tables 9–10: poverty, income, and employment

Read:

- Table 9 shows that the coal decline affects lower-middle poverty areas most strongly, not necessarily the highest-poverty areas.
- Table 10 adds income and employment controls. The coal coefficient remains close to the baseline.
- This suggests that county-level income and employment measures do not absorb the household credit effects.

Economic message: The coal decline appears to affect household finance through broader and more granular channels than county-average income and employment alone.

Table 9
Coal demand interacted with percent of Census block group population below the poverty line in 1999.

	(1) Credit Score	(2) Percent Subprime	(3) Credit Utilization Percent	(4) Delinquent Accounts	(5) Amt 3rd Party Collections ($ 2012)	(6) Percent Bankruptcy
Coal × >0.2 pov share	−0.260	0.131	0.386**	0.00268	9.070	0.0397
	(−0.77)	(0.99)	(2.24)	(0.61)	(1.11)	(1.01)
Coal × 0.1 - 0.2 pov share	−0.501**	0.374***	0.193**	0.00363*	9.933***	0.0584**
	(−2.45)	(4.25)	(2.08)	(1.81)	(2.73)	(2.57)
Coal × <0.1 pov share	−0.377***	0.193**	0.176***	0.00257*	5.279**	0.0383***
	(−2.90)	(2.29)	(3.67)	(1.94)	(2.38)	(3.07)
Mean of Dep. Var.	686.90	38.97	37.75	0.11	193.78	0.47
Observations	1,274,696	1,274,696	935,124	1,293,577	1,313,281	1,392,003
Individuals	183,820	183,820	148,566	178,391	187,575	192,425
Individ w/ Outcome	183,820	96,465	144,686	41,733	79,991	6,741

Source: Census; EIA; NY Fed / Equifax CCP
Notes: Coal demand within 200 miles is interacted with the Census block group's share of population below the poverty line in 1999. Coal tons 200m captures coal (millions of tons) burned within 200 miles of a county centroid by rail. To help with interpretation, coal tons 200m is rescaled by the average annual change between 2011 and 2018. Treatment coefficients are the average one-year change in the outcome variable caused by the coal decline between 2011 and 2018. Observations are at individual-year level and span 2011-2018. Subprime equals 1 when credit score is below 660. Credit utilization percent is based on bankcard and retail trades. Delinquent accounts is the total number of accounts that are 60, 90, 120 or more days past due. Bankruptcy equals 1 when individuals transition into chapter 7 or 13

(a) Table 9: heterogeneity by block-group poverty share

Table 10
Personal income, coal, and credit score.

	(1) Credit Score	(2) Credit Score	(3) Credit Score	(4) Credit Score	(5) Credit Score	(6) Credit Score
Coal tons 200m	−0.364**		−0.367***		−0.353**	
	(−2.61)		(−2.63)		(−2.55)	
Nonfarm personal income per capita (log)	7.836	10.18*				
	(1.31)	(1.70)				
Wage and salary income per capita (log)			4.315**	4.921**		
			(1.99)	(2.21)		
Per capita employment					21.19**	27.32**
					(2.05)	(2.49)
Mean of Dep. Var.	680.64	680.64	680.64	680.64	680.64	680.64
Observations	1,491,301	1,491,301	1,491,301	1,491,301	1,491,301	1,491,301
Individuals	227,109	227,109	227,109	227,109	227,109	227,109
Individ w/ Outcome	227,109	227,109	227,109	227,109	227,109	227,109

(b) Table 10: adding income and employment controls

5.9 Tables 11–12 and Figure 8: migration

Read:

- Table 11 shows that assigning coal demand based on 2010 county residence yields similar results to the baseline.
- Restricting the sample to individuals who do not move counties between 2011 and 2018 also produces similar effects.
- Table 12 shows no significant effect on net migration but a significant decline in churn.
- Figure 8 shows that people who moved out of Appalachian coal counties during 2011–2018 had slower post-move credit-score recovery than people who moved out during 2002–2007.

Economic message: Migration does not explain away the main results. The coal decline seems to harm both places and people, and movers during the decline period appear financially weaker after moving.

Table 11
Sensitivity of results to alternative approaches to treating movers.

(a) Coal tons burned within 200 miles based on counties where individuals lived in 2010

	(1) Credit Score	(2) Percent Subprime	(3) Credit Utilization Percent	(4) Delinquent Accounts	(5) Amt 3rd Party Collections ($ 2012)	(6) Percent Bankruptcy
Coal tons 200m (2010 FIPS)	−0.351***	0.227***	0.210***	0.00241**	8.184***	0.0383***
	(−2.83)	(3.65)	(5.21)	(2.44)	(2.97)	(3.24)
Mean of Dep. Var.	687.12	38.91	37.68	0.11	193.40	0.47
Observations	1,274,527	1,274,527	935,995	1,293,231	1,313,168	1,391,396
Individuals	181,045	181,045	146,997	175,496	184,658	189,418
Individ w/ Outcome	181,045	94,547	143,114	40,683	78,326	6,576

(b) Sample restricted to individuals who did not move counties between 2011 and 2018

	(1) Credit Score	(2) Percent Subprime	(3) Credit Utilization Percent	(4) Delinquent Accounts	(5) Amt 3rd Party Collections ($ 2012)	(6) Percent Bankruptcy
Coal tons 200m	−0.317**	0.195***	0.221***	0.00212**	9.545***	0.0495***
	(−2.21)	(3.12)	(4.96)	(2.08)	(2.78)	(3.94)
Mean of Dep. Var.	686.23	38.81	37.07	0.10	188.60	0.44
Observations	1,058,300	1,058,300	763,044	1,070,996	1,094,412	1,136,814
Individuals	159,931	159,931	125,216	154,174	163,681	165,062
Individ w/ Outcome	159,931	83,776	121,349	33,896	67,960	4,950

Source: EIA; NY Fed / Equifax CCP
Note: Coal tons 200m (2010 FIPS) in Panel A captures millions of coal tons burned within 200 miles of a county centroid by rail, where the county centroid is based on the individual's county of residence in 2010. Coal tons 200m in Panel B captures millions of coal tons burned within 200 miles of a county centroid by rail, where the sample is restricted to individuals who did not change counties of residence from 2011 to 2018. To help with interpretation, coal tons 200m is rescaled by the average annual change between 2011 and 2018. Treatment coefficients are the average one-year change in the outcome variable caused by the coal decline between 2011 and 2018. Observations are at individual-year level and span 2011-2018. Subprime equals 1 when credit score is below 660. Credit utilization percent is based on bankcard and retail trades. Delinquent accounts is the total number of accounts that are 60, 90, 120 or more days past due. Bankruptcy equals 1 when individuals transition into

Figure 13: Table 11: sensitivity to alternative approaches to treating movers

Table 12
County-level regression for churn and migration.

	(1) Net Migration (Percent)	(2) Churn (Percent)
Coal tons 200m	−0.00252	−0.160**
	(−0.03)	(−2.07)
Mean of Dep. Var.	-0.14	5.20
Observations	976	976
Number of Counties	122	122

Source: Census; EIA; NY Fed / Equifax CCP
Notes: Coal tons 200m captures millions of coal tons burned within 200 miles c a county centroid by rail. To help with interpretation, coal tons 200m is rescale by the average annual change between 2011 and 2018. Treatment coefficient are the average one-year change in the outcome variable caused by the coa decline between 2011 and 2018. For this table, an individual is considered t have moved if they are in one county for at least four consecutive quarter

(a) Table 12: county-level churn and migration

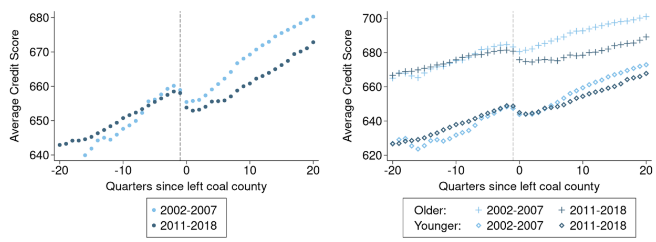


Fig. 8. Credit scores of people who moved out of Appalachia
Note: The points in this chart correspond to the average credit score based on the quarter in relation to when the individual left coal counties. The qu 0 is the first quarter in which the individual is no longer in an Appalachian coal mining county. The vertical dashed lines indicate quarter −1, the las the move. For panel (b), age is categorized based on the age of the person the year they move, with individuals who were 45 and younger at designated as "younger." For both panels, the sample only includes individuals who lived in active coal mining counties for at least four consecutive qu made one move out of the region. The panels exclude individuals whose data indicated they moved during quarters with anomalously large nu (primarily in the early sample). The timing of these moves was unlikely to be accurate and may introduce significant noise.

(b) Figure 8: credit scores of people who moved out of Appalachia

6 Short summary

- The paper studies the household-finance consequences of the Appalachian coal decline between 2011 and 2018.
- The main dataset is the NY Fed/Equifax Consumer Credit Panel, linked to county-level coal mining and electricity-sector demand data.
- The key treatment is coal consumed at power plants within 200 miles of a county by rail.
- The shock is motivated by cheap natural gas and preexisting gas generation capacity, which shifted electricity generation away from coal.
- The baseline regression includes individual fixed effects, county fixed effects, year fixed effects, and individual time trends.
- Coal-demand declines predict lower local coal production, mining hours, and coal employment.
- The average annual coal decline lowers credit scores by about 0.386 points and raises several distress outcomes.
- Over the seven-year sample, the implied average credit-score decline is nearly 3 points.
- Effects materialize quickly, generally within two years.
- The largest credit-score effects are near the subprime threshold.
- Older cohorts, especially Baby Boomers and Generation X households, are more affected than Millennials.
- Direct coal-worker job losses cannot fully explain the broad credit-score deterioration.
- Migration does not drive the results, and net migration does not respond significantly.
- The paper's policy message is that fossil-fuel transitions can create fast, broad, and distributionally uneven household-finance costs.

7 Conclusion

7.1 Core conclusion

Blonz, Roth Tran, and Troland show that the decline in Appalachian coal demand harmed household finances in coal-mining communities. Credit scores fell, subprime status rose, credit utilization increased, delinquency rose, third-party collections increased, and bankruptcies became more likely. These effects appeared within two years and were not limited to direct coal-worker households.

7.2 Economic intuition

The economic mechanism is local demand spillover. Cheap natural gas reduced electricity-sector demand for Appalachian coal. That demand shock reduced coal activity and weakened local economic conditions. Households responded through worse credit outcomes, either because of direct job and income losses, local spillovers, reduced opportunities, or greater household fragility.

7.3 Takeaway

The paper turns the fossil-fuel transition from an abstract macro policy issue into a household balance-sheet issue. Even when direct extraction employment is a small share of local employment, the decline of a legacy industry can quickly weaken credit health in the surrounding community. For policy, the key lesson is that transition costs may be broad, fast, and concentrated among households whose credit access is already fragile.